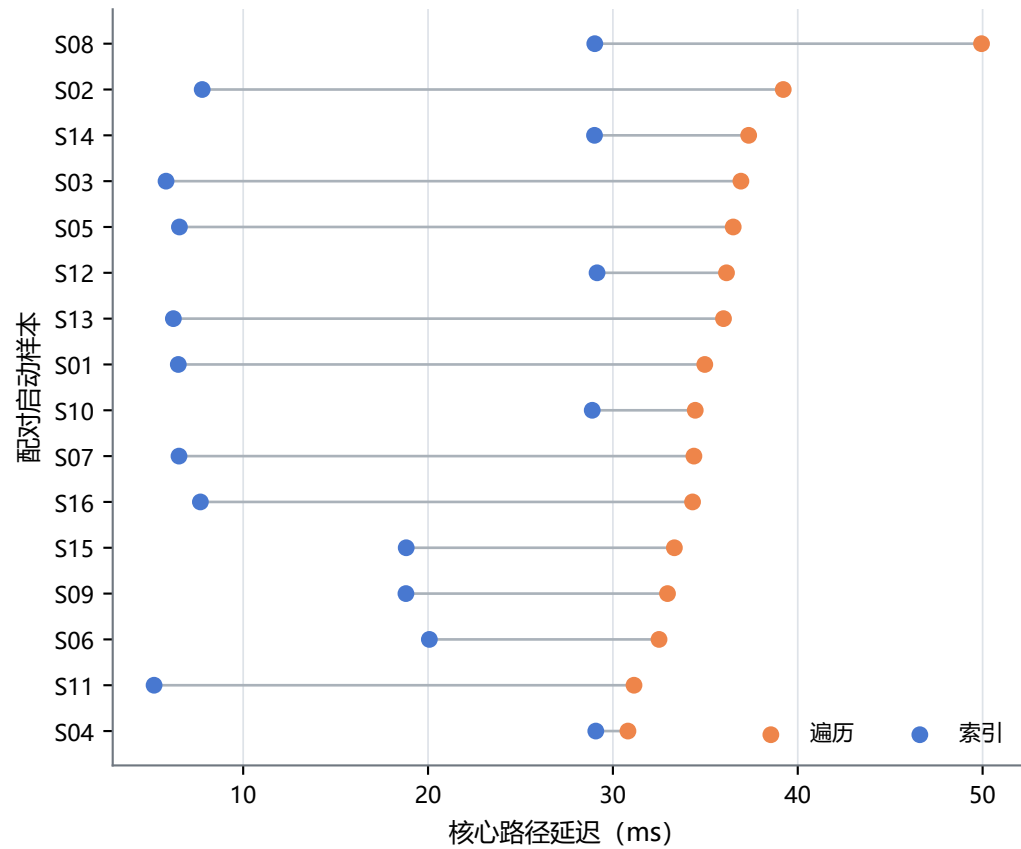


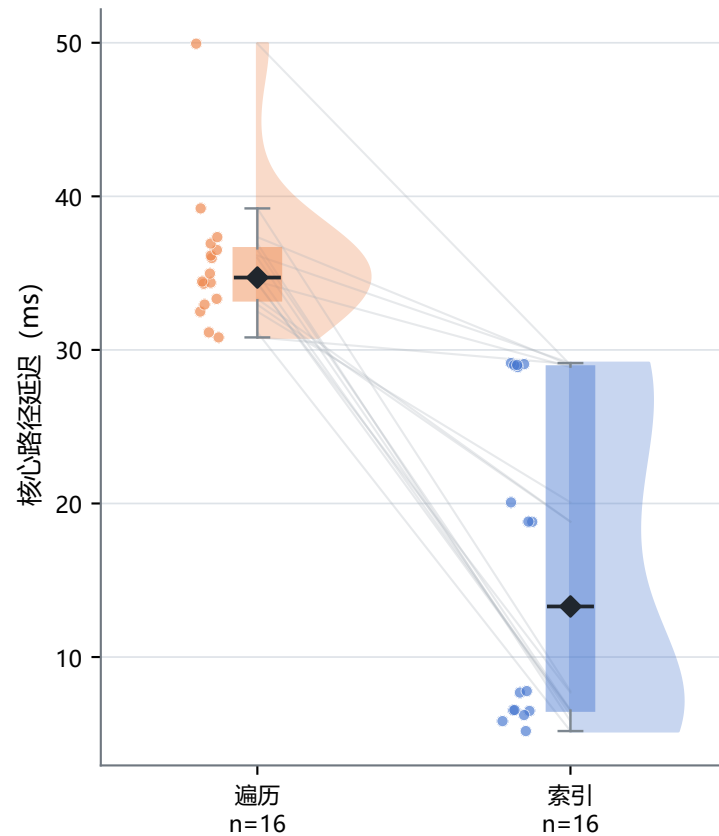
a

逐启动配对哑铃图



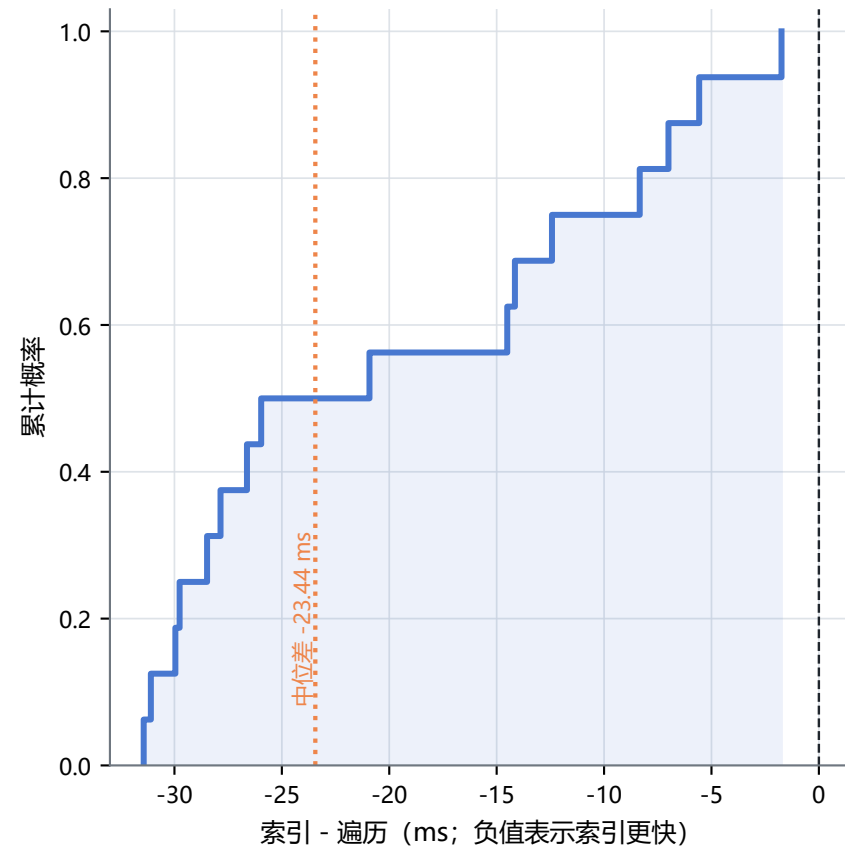
b

配对雨云分布



c

配对差值 ECDF



数据: contest\_paired.csv; n=16 对 fresh boot; 索引核心路径更快 16/16 对; 配对加速比中位数 3.118x。横线连接同一启动中的两个路径。